Prob. 1: Choose the correct answer.

		A	B	C	D	Your answer
	Consider the circuit shown in Fig. 1 where V = 15 V, and I = 0.75 A.					
1	The voltage V_{ab} (in V) due to the current source (I) only is	(9)	12	6	3	
2	The voltage V_{ab} (in V) due to the voltage source (V) only is	3	6	12	(9)	
3	The voltage V_{ab} (in V) due to both sources is	24	6	(18)	12	
	Consider the circuit shown in Fig. 2 where I_s = 2 A, R_1 = 7.5 Ω.					
4	The equivalent resistance (R_{eq}) between a–b (in Ω) is	(6)	10	12	8	
5	The power supplied (in W) by the current source is	32	40	48	(24)	

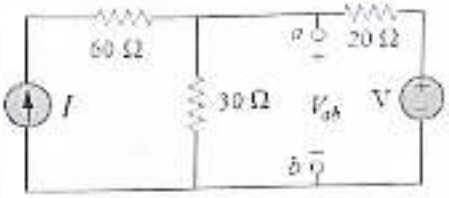


Figure 1

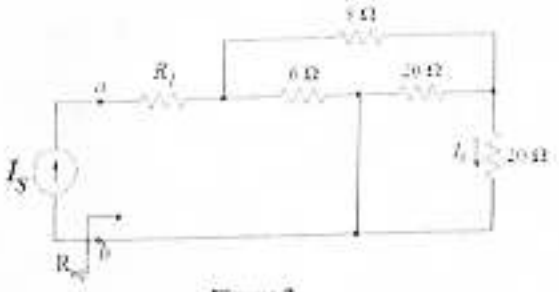


Figure 2

Prob. 2: (a) The charge transferred through an element is given by $q(t)=3t^3+6t^2+5t$ C. Calculate the current entering the element at instant $t = 5$ sec.

$i(t) = \frac{dq}{dt} = 9t^2 + 12t + 5 \Rightarrow i(5) = 2\,a\,a$ A

(b) A bulb light draws 280W when connected to a 120 V line. Calculate the conductance of the bulb.

$P = V \cdot i \Rightarrow 280 = 120 \cdot i \Rightarrow i = 2.33$ A

$R = \frac{120}{2.33} = 51.82\ \Omega$

$G = 0.019455$

Prob. 3: Consider the circuit shown in Fig. 3 where I_s = 1A and V_s = 8V. Using the Nodal method

a. Write all the required equations to find V_1, V_2, and i_x (***Do not simplify***)

.....

.....

.....

b. Solve to find V_1 = 17.33 V, V_2 = 16 V, and i_x = 0.133 A

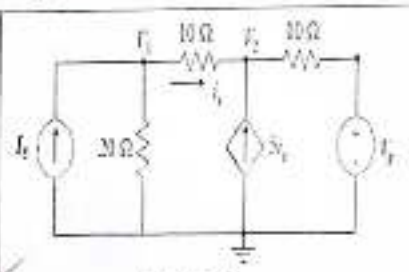


Figure 3

Prob. 4: Consider the circuit shown in Fig. 4 where I_s = 6A, V_s = 12V, and R_1 = 7 Ω. Then:

a. The open circuit voltage ($V_{oc} = V_{ab}$) between the a–b terminals is .....

b. The Thevenin equivalent resistance (R_{th}) between the a–b terminals is 2.1 Ω

c. The current of a load of 20 Ω that is connected between the a–b terminals is .....

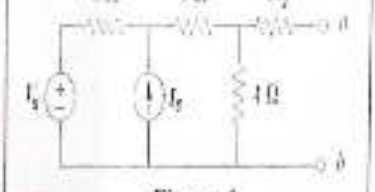


Figure 4

Prob. 1: Choose the correct answer.

		A	B	C	D	Your answer
	Consider the circuit shown in Fig. 1 where V = 10 V, and I = 0.5 A.					
1	The voltage V_{ab} (in V) due to the current source (I) only is	9	12	6	3	
2	The voltage V_{ab} (in V) due to the voltage source (V) only is	3	6	12	9	
3	The voltage V_{ab} (in V) due to both sources is	24	6	18	12	
	Consider the circuit shown in Fig. 2 where I_S = 2 A, R_1 = 5.5 Ω.					
4	The equivalent resistance (R_{eq}) between a–b (in Ω) is	6	10	12	8	
5	The power supplied (in W) by the current source is	32	40	48	24	

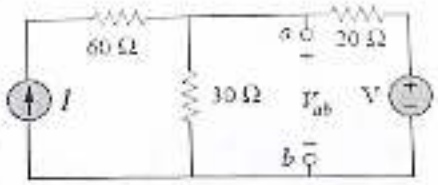


Figure 1

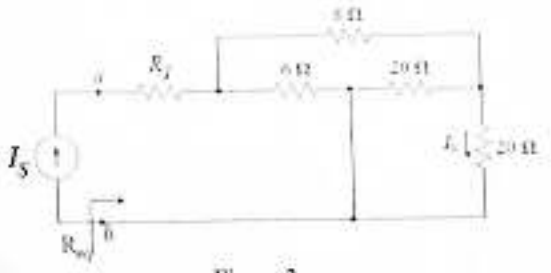


Figure 2

Prob. 2: (a) The charge transferred through an element is given by $q(t)=3t^3+4t^2+5t$ C. Calculate the current entering the element at instant $t = 5$ sec.

$i = \frac{dq}{dt} \Rightarrow i = 9t^2 + 8t + 5$

$i = 9(25) + 8(5) + 5$

$i = 270$ A

(b) A bulb light draws 240W when connected to a 120 V line. Calculate the conductance of the bulb.

$P = Vi$

$P = \frac{V^2}{R} = V^2 G \Rightarrow 240 = (120)^2 G \Rightarrow G = \frac{1}{60}$

Prob. 3: Consider the circuit shown in Fig. 3 where I_S = 3A and V_S = 12V. Using the Nodal method

a. Write all the required equations to find V_1, V_2, and i_x (***Do not simplify***)

$I_S = I_1 + i_x$ $\quad$ $i_x + 5i_x + I_{VS} = 0$ $\quad$ $(6i_x + I_{VS} = 0)$

$3 = \frac{V_1}{20} + \frac{V_1 - V_2}{10}$ $\quad$ $6\left(\frac{V_1 - V_2}{10}\right) + \frac{V_2 - 12}{10} = 0$

b. Solve to find V_1 = 92 V, V_2 = 108 V, and i_x = −1.6 A.

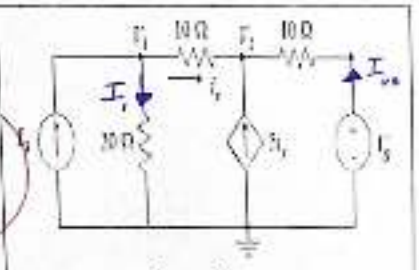


Figure 3

Prob. 4: Consider the circuit shown in Fig. 4 where I_S = 4A, V_S = 16V, and R_1 = 5 Ω. Then:

a. The open circuit voltage ($V_{ab} = V_{oc}$) between the a–b terminals is 10 V

b. The Thevenin equivalent resistance (R_{th}) between the a–b terminals is 8 Ω

c. The current of a load of 20 Ω that is connected between the a–b terminals is $\frac{5}{14}$

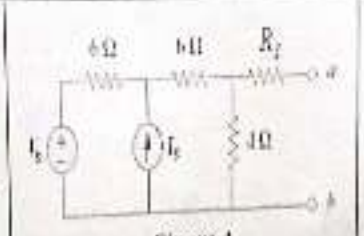


Figure 4

Prob. 1: Choose the correct answer.

		A	B	C	D	Your answer
	Consider the circuit shown in Fig. 1 where $V = 20$ V, and $I = 1.6$ A.					
1	The voltage V_{ab} (in V) due to the current source (I) only is	9	12	6	3	B
2	The voltage V_{ab} (in V) due to the voltage source (V) only is	3	6	12	9	C
3	The voltage V_{ab} (in V) due to both sources is	24	6	18	12	A
	Consider the circuit shown in Fig. 2 where $I_S = 2$ A, $R_1 = 1.5\ \Omega$.					
4	The equivalent resistance (R_{eq}) between a–b (in Ω) is	4	10	12	8	A
5	The power supplied (in W) by the current source is	32	40	48	24	D

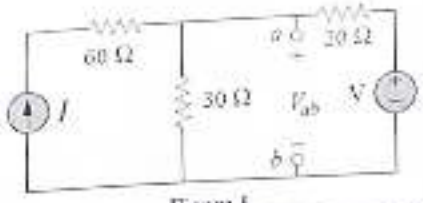


Figure 1

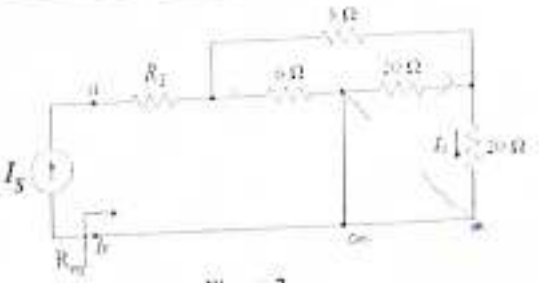


Figure 2

Prob. 2: (a) The charge transferred through an element is given by $q(t) = 3t^3 + 8t^2 + 5t$ C. Calculate the current entering the element at instant $t = 5$ sec.

$i = \frac{dq}{dt} = q(5) = 9t^2 + 16t + 5 = 310$ A

(b) A bulb light draws 320W when connected to a 120 V line. Calculate the conductance of the bulb.

$P = \frac{V^2}{R}$ $320 = \frac{120^2}{R} \Rightarrow 320R = 14400$

$R = 45$

$0.022 = \frac{1}{45} = \frac{1}{R}$ Ω

Prob. 3: Consider the circuit shown in Fig. 3 where $I_S = 4$A and $V_S = 15$V.
Using the Nodal method

a. Write all the required equations to find V_1, V_2, and i_x (*Do not simplify*)

$\frac{V_1 - V_2}{10} + \frac{V_1}{20} = 4$ ---(1)

$\frac{V_2 - V_1}{10} + \frac{V_2}{10} = 5i_x$ ---(2)

$\frac{V_1 - V_2}{10} = i_x$

b. Solve to find V_1 = 65.5 V, V_2 = 58.3 V, and i_x = 0.72 A

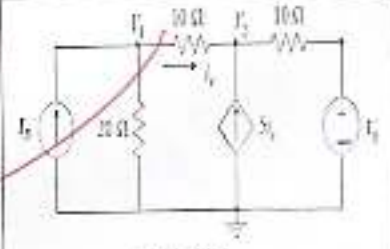


Figure 3

Prob. 4: Consider the circuit shown in Fig. 4 where $I_S = 8$A, $V_S = 8$V, and $R_1 = 9\ \Omega$. Then:

a. The open circuit voltage ($V_{ab} = V_a$) between the a–b terminals is 14

b. The Thevenin equivalent resistance (R_{th}) between the a–b terminals is 16

c. The current of a load of 20 Ω that is connected between the a–b terminals is 0.527

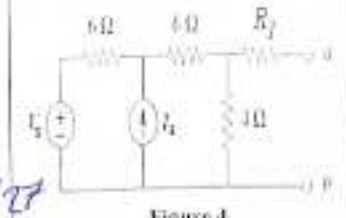


Figure 4